

第四章 振动

- 简谐振动
- 简谐振动的合成



第一节 简谐振动

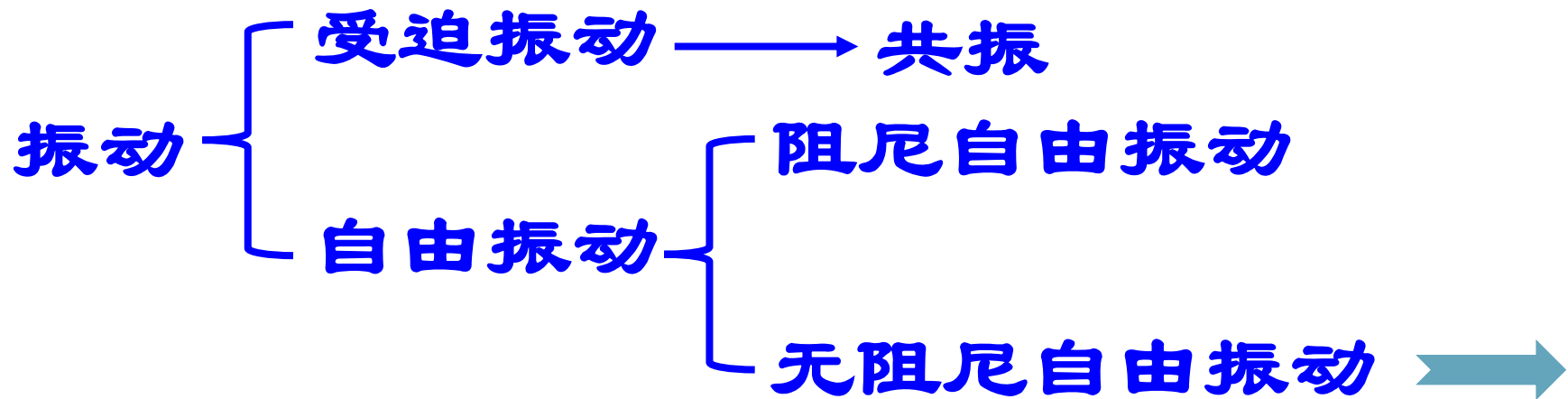
机械振动

物体在某一位置附近来回往复的运动

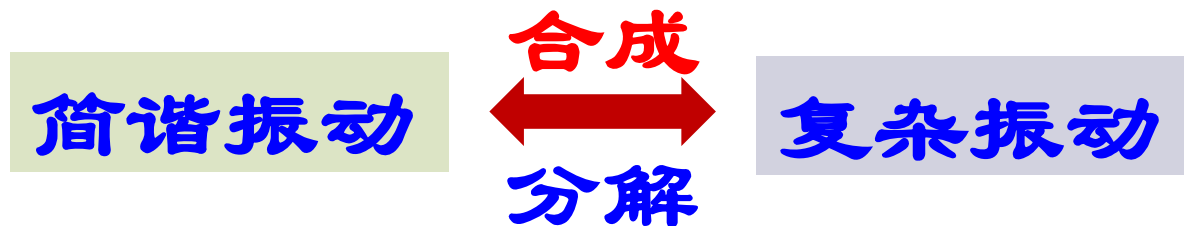
广义振动

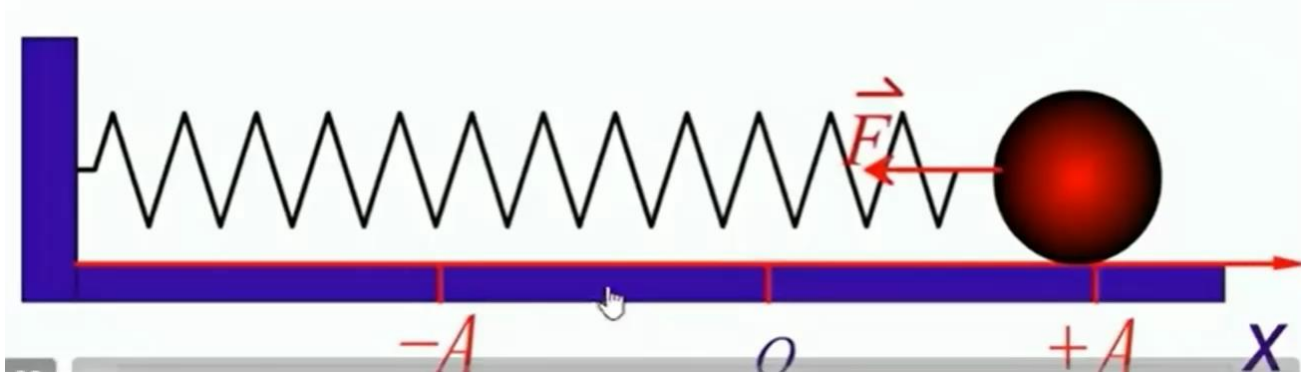
任何一个物理量在某一量值附近随时间作周期性变化都可以叫做振动

例如：交流电路中的电流、电压，无线电传播时，空间某点的电场强度和磁场强度

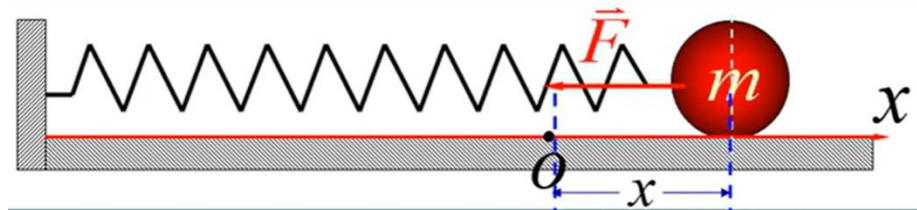
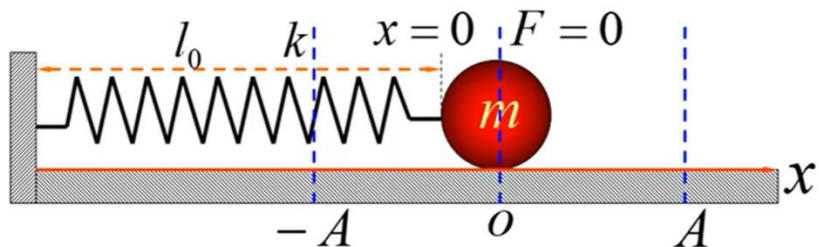


简谐振动 最简单、最基本的振动





弹簧振子：理想模型 **0：平衡位置**



$$F = -kx \quad \text{回复力} \quad F = ma \quad \longrightarrow$$

$$-kx = ma \quad a = -(k / m) x \quad a = -\omega^2 x$$

令 $\omega^2 = k / m$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$\frac{d^2 x}{dt^2} = -\omega^2 x$$

简谐振动的动力学方程

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$x = A \cos(\omega t + \varphi)$$

积分常数，根据初始条件确定

简谐振动运动学方程

特点： (1) 等幅振动 (2) 周期性振动

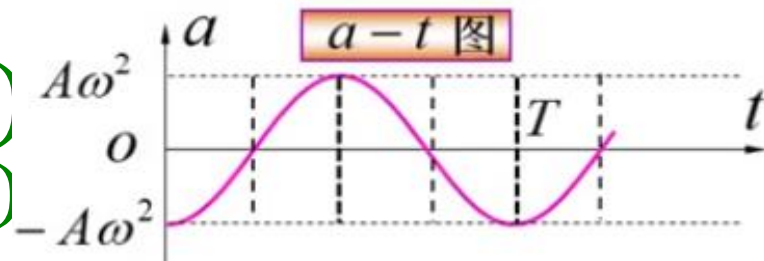
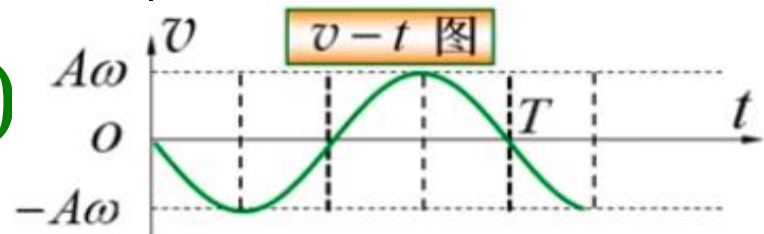
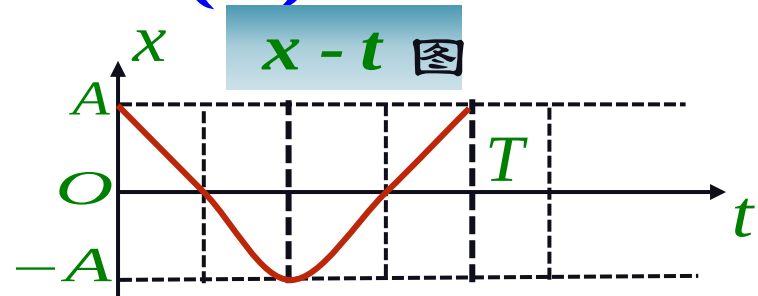
$$x = A \cos(\omega t + \varphi)$$

$$v = -A\omega \sin(\omega t + \varphi)$$

$$v = A\omega \cos(\omega t + \varphi + \frac{\pi}{2})$$

$$a = -A\omega^2 \cos(\omega t + \varphi)$$

$$a = A\omega^2 \cos(\omega t + \varphi + \pi)$$



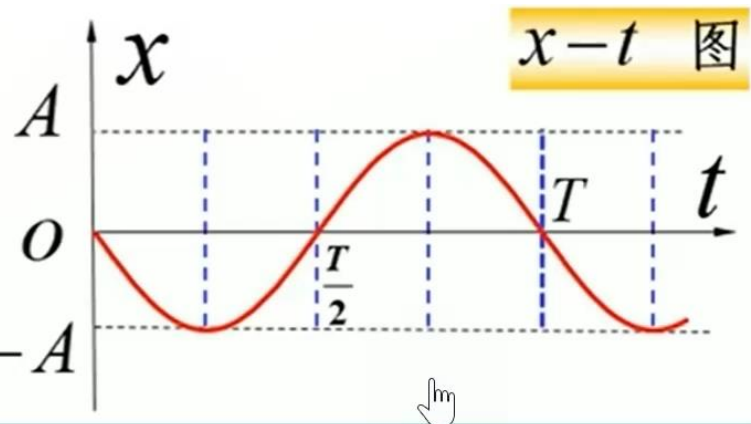
简谐振动的特征量

(1) 振幅 $A = |x_{\max}|$

(2) 周期、频率 周期

频率 $\nu = 1/T = \omega / 2\pi$

圆频率 $\omega = 2\pi\nu = 2\pi / T$



弹簧振子: $\omega = \sqrt{k/m}$

$T = 2\pi\sqrt{m/k}$

周期和频率仅与振动系统本身的物理性质有关

(3) 相位 $\omega t + \varphi$

a) $\omega t + \varphi \rightarrow (x, v)$ 存在一一对应关系

b) 相位在 $0 \rightarrow 2\pi$ 内变化时, 物体无相同的运动状态

c) 初相位 $\varphi(t=0)$ 描述物体初始时刻

的运动状态, (φ 取 $[-\pi \rightarrow \pi]$ 或 $[0 \rightarrow 2\pi]$)

常数 A 和 φ 的确定

$$\begin{cases} x = A \cos(\omega t + \varphi) \\ v = -A\omega \sin(\omega t + \varphi) \end{cases}$$

设 初始条件: $t = 0$ 时, $x = x_0, v = v_0$

$$\begin{cases} x_0 = A \cos \varphi \\ v_0 = -A\omega \sin \varphi \end{cases}$$



$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}}$$

$$\tan \varphi = \frac{-v_0}{\omega x_0}$$

例题：一弹簧振子的质量为 m ，劲度系数为 k 。 $t = 0$ 时， $x = x_0$ ， $v = v_0$ ，且 $v_0 < 0$ 求该谐振子的运动学方程。

解：

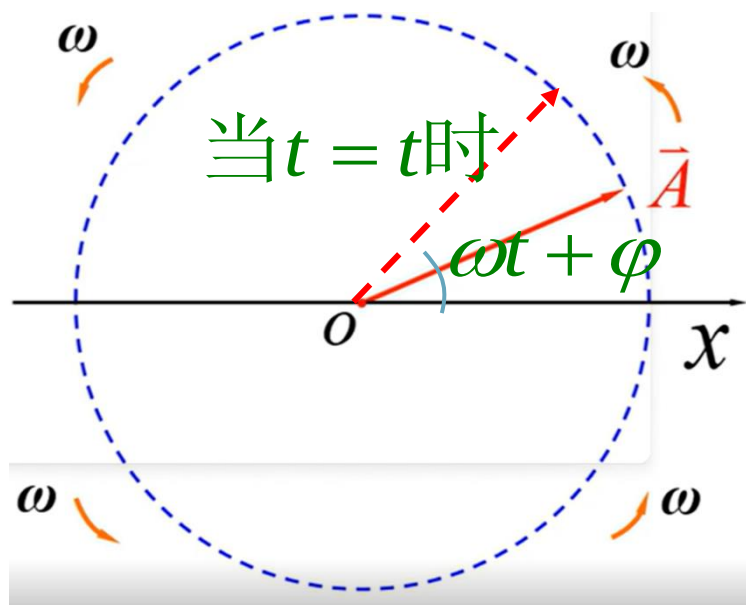
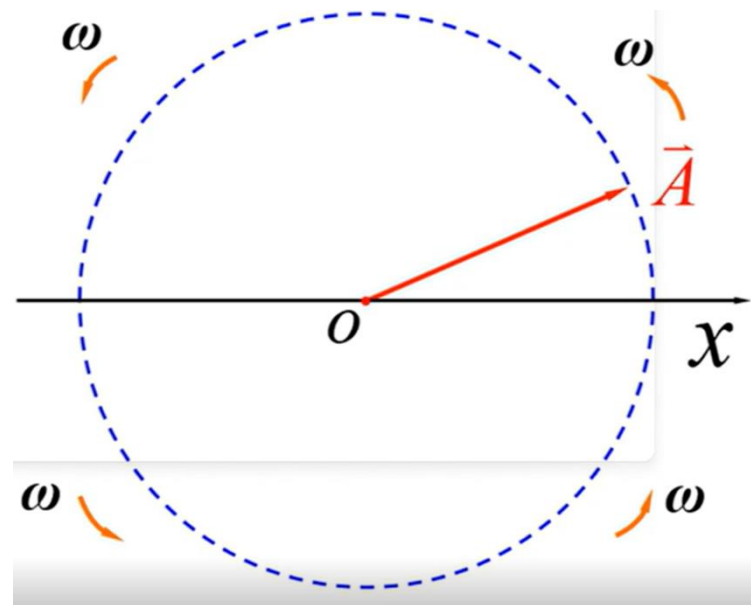
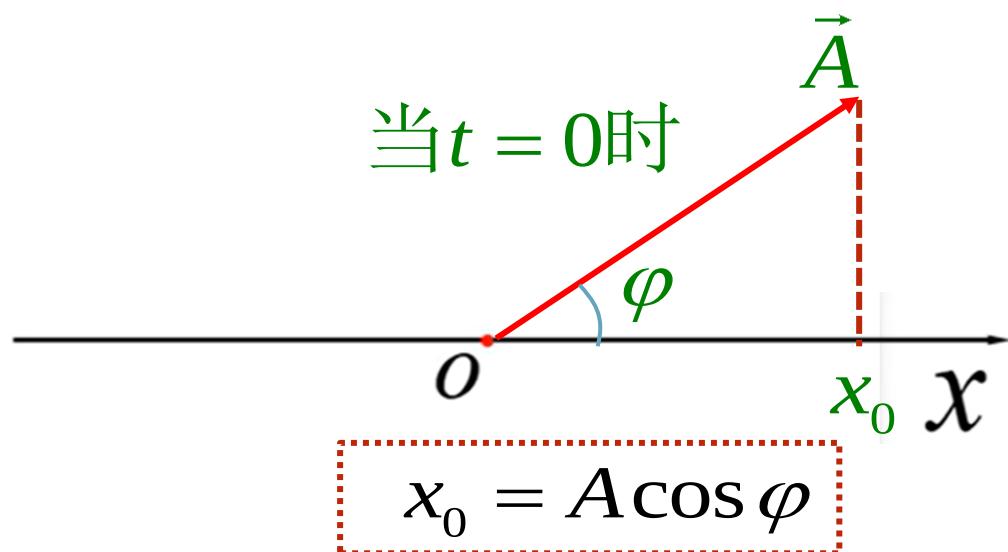
$$\omega = \sqrt{k/m} \quad A = v_0 \sqrt{m/k} \quad \tan \varphi = \infty \quad \varphi = \pm \pi/2$$

$$v = -A\omega \sin(\omega t + \varphi) \quad v_0 = -A\omega \sin \varphi < 0$$

$$\sin \varphi > 0 \quad \varphi = \pi/2$$

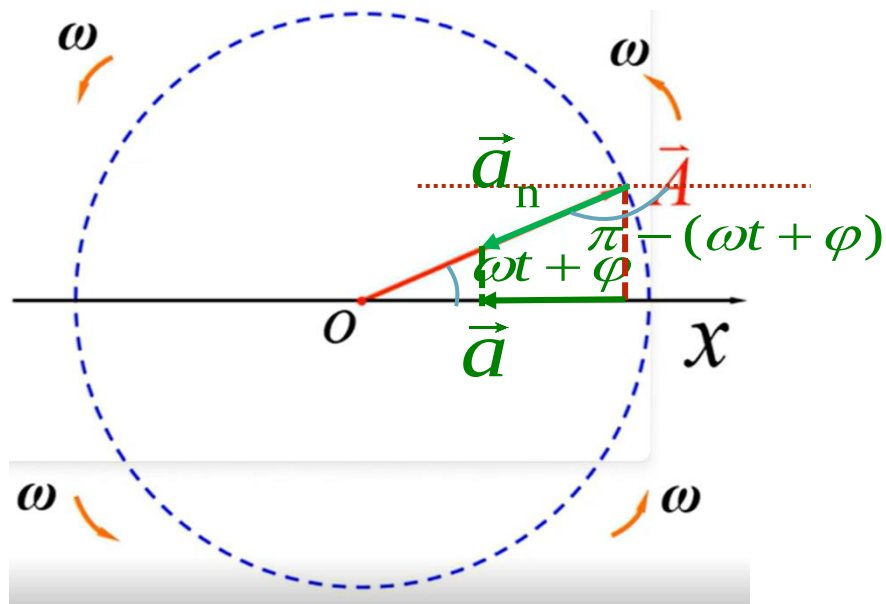
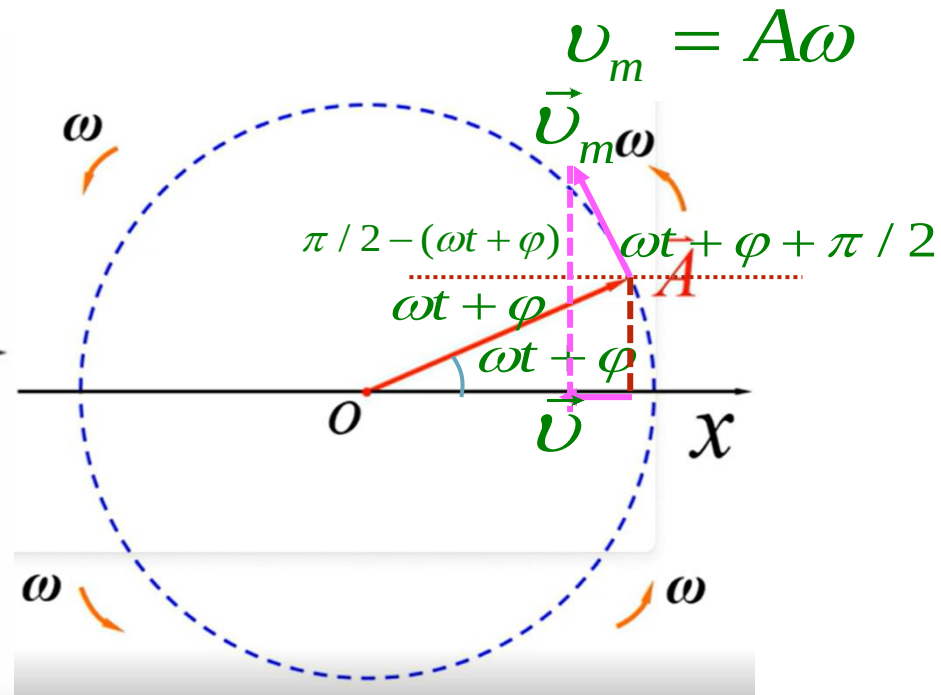
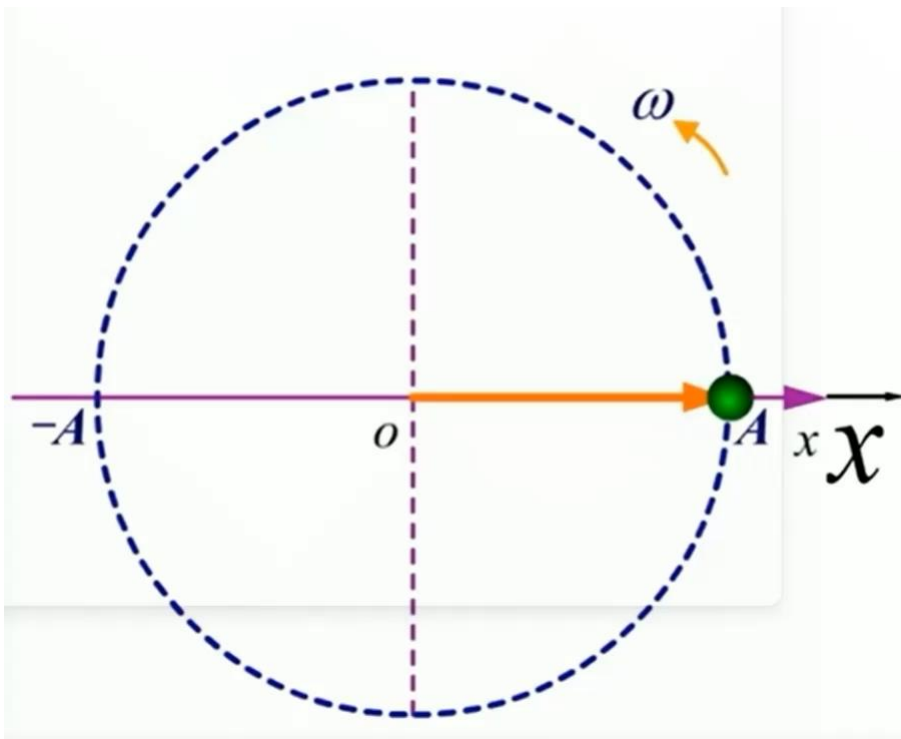
$$\Rightarrow x = v_0 \sqrt{\frac{m}{k}} \cos\left(\sqrt{\frac{k}{m}}t + \frac{\pi}{2}\right)$$

旋转矢量表示法



$$x = A \cos(\omega t + \varphi)$$

一个以 ω 角速度逆时针旋转的矢量 $\vec{A}$ ，其端点在 x 轴上的投影点的运动就是简谐振动



$$v = A\omega \cos(\omega t + \varphi + \pi/2)$$

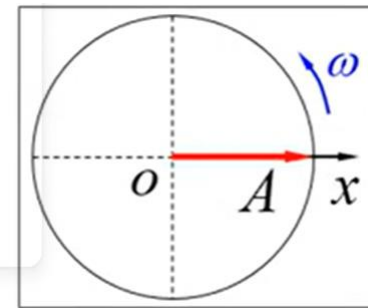
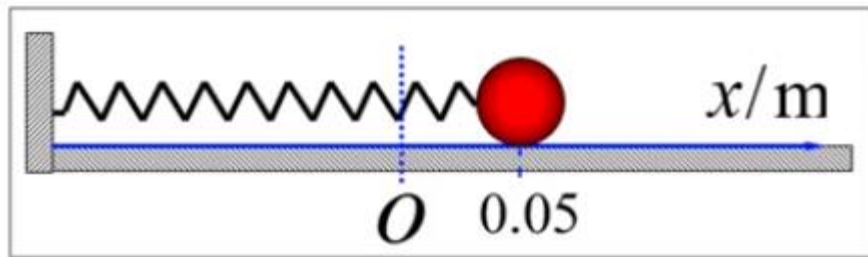
$$= -A\sin(\omega t + \varphi)$$

$$a_m = A\omega^2$$

$$a = A\omega^2 \cos[\pi - (\omega t + \varphi)]$$

$$= -A\omega^2 \cos(\omega t + \varphi)$$

例题：如图所示，轻弹簧的右端连着一物体，弹簧劲度系数 $k = 0.72 \text{ N} \cdot \text{m}^{-1}$ ，物体质量为 $m = 20 \text{ g}$ ，（1）把物体从平衡位置向右拉到 $x = 0.05 \text{ m}$ 处停下后再释放，求其简谐振动方程



解： $\omega = \sqrt{k / m} = \sqrt{0.72 \text{ N} \cdot \text{m}^{-1} / 0.02 \text{ kg}} = 6 \text{ s}^{-1}$

$$A = \sqrt{x_0^2 + v_0^2 / \omega^2} = x_0 = 0.05 \text{ m} \quad \tan \varphi = -v_0 / \omega x_0 = 0$$

$$\varphi = 0 \text{ 或 } \pi$$

由旋转矢量图可知 $\varphi = 0$

$$x = A \cos(\omega t + \varphi) = (0.05 \text{ m}) \cos \left[(6 \text{ s}^{-1}) t \right]$$

(2) 求物体从初位置运动到第1次经过 $A/2$ 处时的速度

$$v = -A\omega \sin(\omega t + \varphi)$$

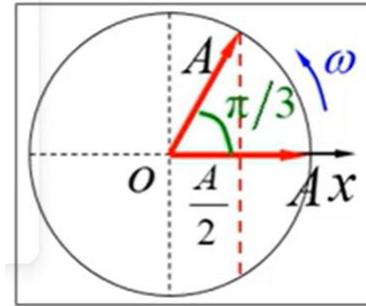
$$= -A\omega \sin \omega t$$

$$x = A \cos \omega t$$

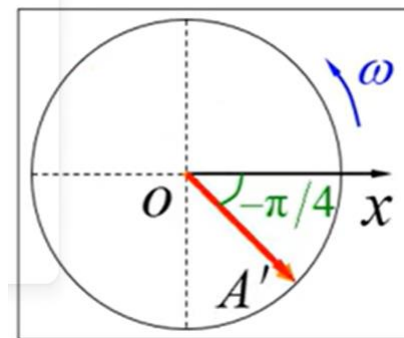
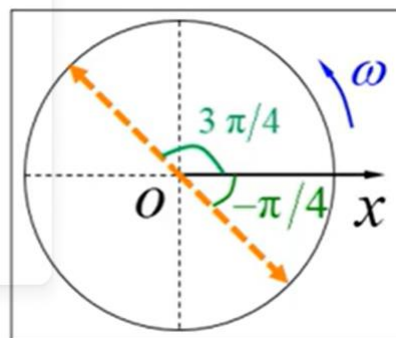
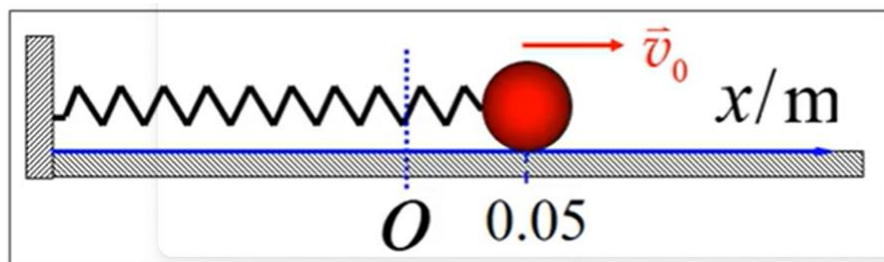
$$\cos \omega t = x / A = 1/2 \quad \longrightarrow \quad \omega t = \pi / 3 \text{ 或 } 5\pi / 3$$

由旋转矢量图可知 $\omega t = \pi / 3$

$$v = -A\omega \sin(\omega t + \varphi) = 0.26 \text{m} \cdot \text{s}^{-1}$$



(3) 如果物体在 $x=0.05\text{m}$ 处时速度不等于零，而是具有向右的初速度 $v_0 = 0.3\text{m} \cdot \text{s}^{-1}$ ，求其运动方程。



$$A' = \sqrt{x_0^2 + v_0^2 / \omega^2} = 0.0707\text{m}$$

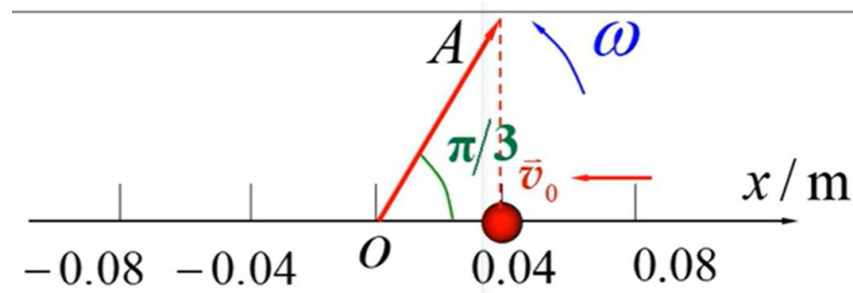
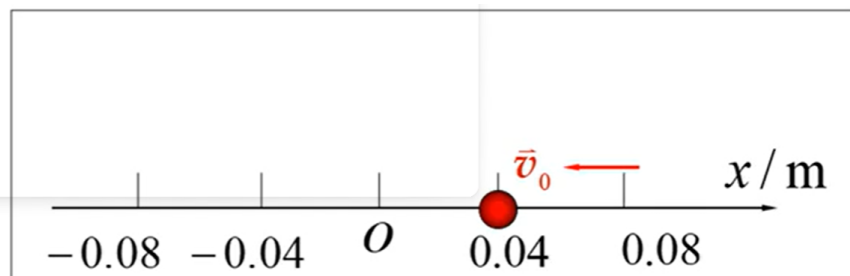
$$\tan \varphi' = -v_0 / \omega x_0 = -1 \quad \varphi' = -\pi / 4 \text{ 或 } 3\pi / 4$$

由旋转矢量图：

$$\varphi' = -\pi / 4$$

$$x = A \cos(\omega t + \varphi) = (0.0707\text{m}) \cos \left[(6\text{s}^{-1})t - \pi / 4 \right]$$

例题：质量为 0.01kg 的物体做简谐运动，振幅为 0.08m ，周期为 4s ，起始时刻物体在 $x=0.04\text{m}$ ，向 Ox 轴负方向运动（如图）。试求：（1） $t=1\text{s}$ 时，物体所处的位置和所受的力，（2）由起始位置运动到 $x=-0.04\text{m}$ 处所需要的最短时间。



解：(1) $A = 0.08\text{m}$ $T = 4\text{s}$ $\omega = 2\pi / T = (\pi/2) \text{ s}^{-1}$

$t = 0, x = 0.04\text{m}$ 代入 $x = A\cos(\omega t + \varphi)$ $0.04\text{m} = (0.08\text{m})\cos\varphi$

$\varphi = \pm\pi/3$ 因为： $v_0 < 0$ ，所以 $\varphi = \pi/3$

$$x = (0.08\text{m})\cos\left[(\pi/2\text{s}^{-1})t + \pi/3\right] \quad t = 1\text{s} \quad x = -0.069\text{m}$$

$$F = -kx = -m\omega^2 x = -(0.01\text{kg}) \times (\pi / 2\text{s}^{-1})^2 \times (-0.069\text{m}) = 1.70 \times 10^{-3}$$

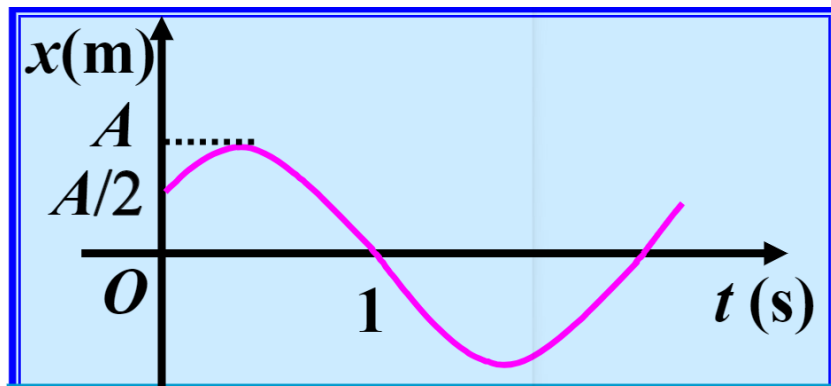
(2) 设由起始位置运动到 $x = -0.04\text{m}$ 处所需要的最短时间为 t

$$x = (0.08\text{m}) \cos \left[(\pi / 2\text{s}^{-1})t + \pi / 3 \right]$$

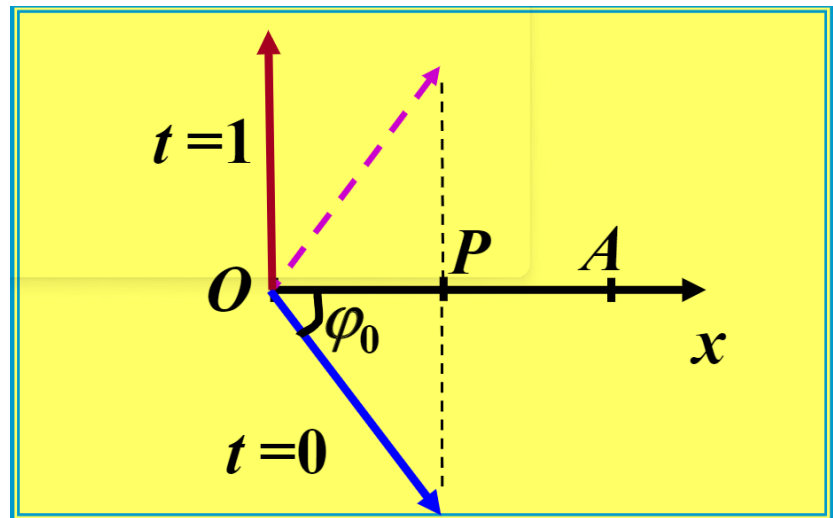
$$-0.04\text{m} = (0.08\text{m}) \cos \left[(\pi / 2\text{s}^{-1})t + \pi / 3 \right]$$

$$t = \left\{ \left[\arccos(-1/2) - \pi / 3 \right] / \pi / 2 \right\} = 2 / 3\text{s} = 0.667\text{s}$$

例题：已知简谐振动曲线如图所示，求其振动方程。



解： $t = 0 \quad x = A/2 \quad \varphi_0 = -\pi/3$
 $t = 1\text{s} \quad x = 0 \quad \varphi_1 = \pi/2$
 $\varphi_1 = \omega t + \varphi_0$
 $\pi/2 = \omega \times 1 + (-\pi/3) \quad \omega = \frac{5\pi}{6}$
$$x = A \cos\left(\frac{5\pi}{6}t - \frac{\pi}{3}\right)$$



简谐振动能量

动能

$$v = -A\omega \sin(\omega t + \varphi)$$

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}mA^2\omega^2 \sin^2(\omega t + \varphi)$$

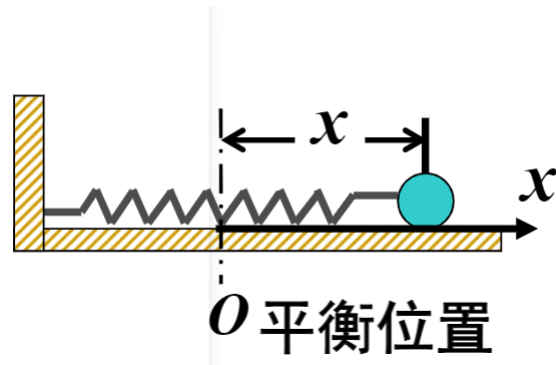
势能

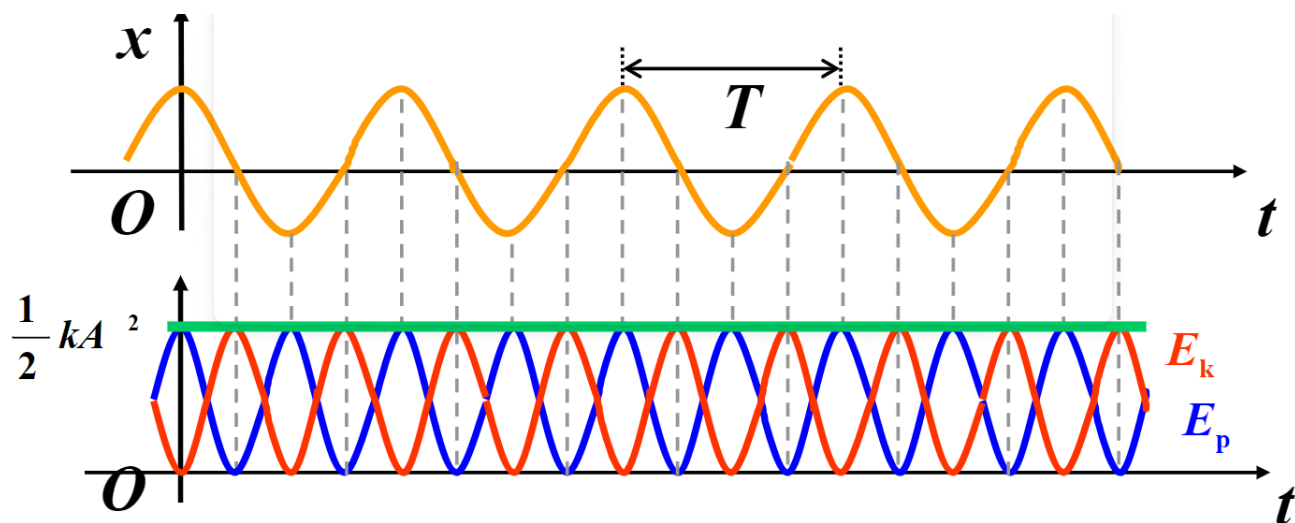
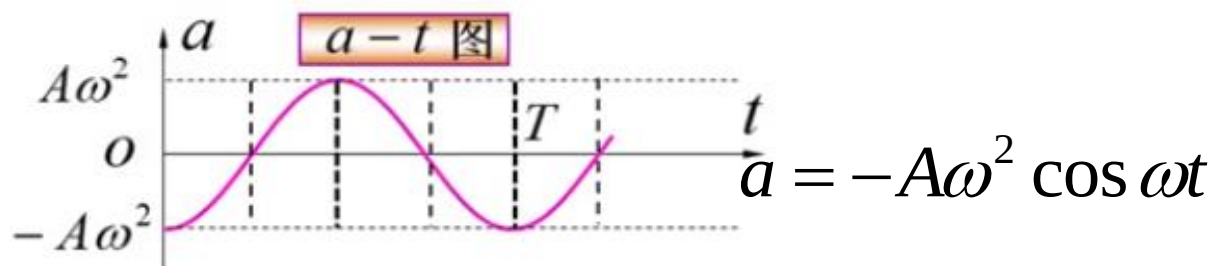
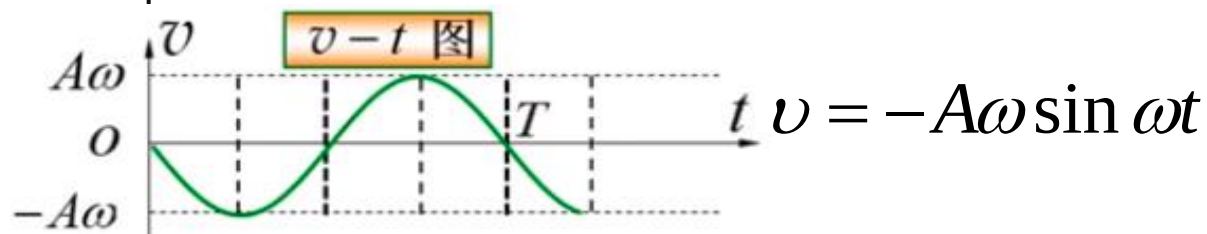
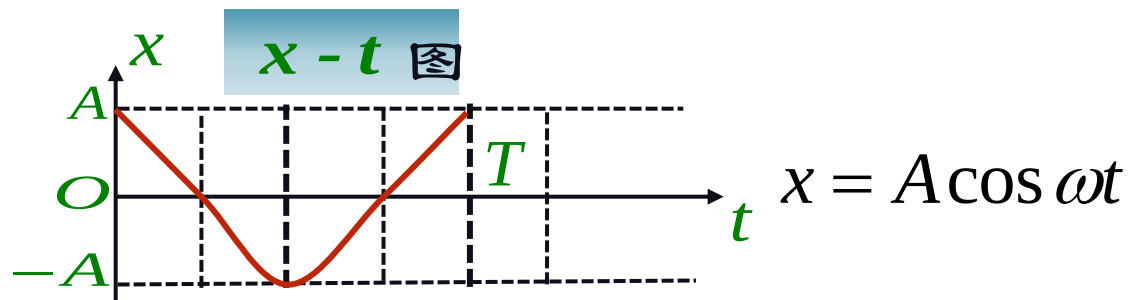
$$x = A\cos(\omega t + \varphi)$$

$$E_p = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega t + \varphi)$$

机械能

$$E = E_k + E_p = \frac{1}{2}mA^2\omega^2 = \frac{1}{2}kA^2$$





简谐振动的合成

1、两个同方向、同频率简谐振动的合成

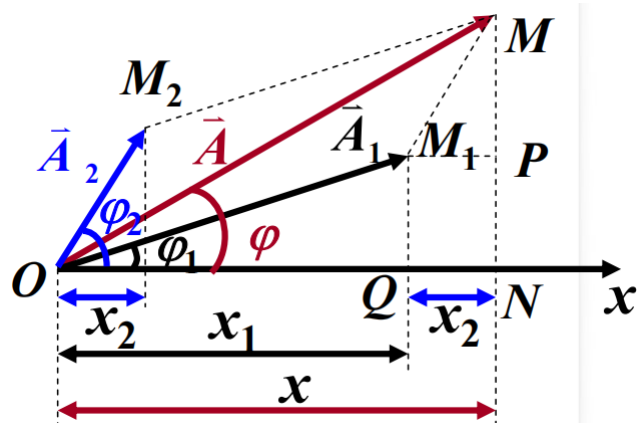
$$x_1 = A_1 \cos(\omega t + \varphi_1)$$

$$X = X_1 + X_2$$

$$x_2 = A_2 \cos(\omega t + \varphi_2)$$

$$x = A \cos(\omega t + \varphi)$$

两个同方向同频率简谐振动合成后仍为简谐振动



$$A^2 = A_1^2 + A_2^2 - 2A_1A_2 \cos[\pi - (\varphi_2 - \varphi_1)]$$

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

$$(1) A_2 \sin \varphi_2$$

$$(2) A_1 \sin \varphi_1$$

$$(3) A_2 \cos \varphi_2$$

$$(4) A_1 \cos \varphi_1$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

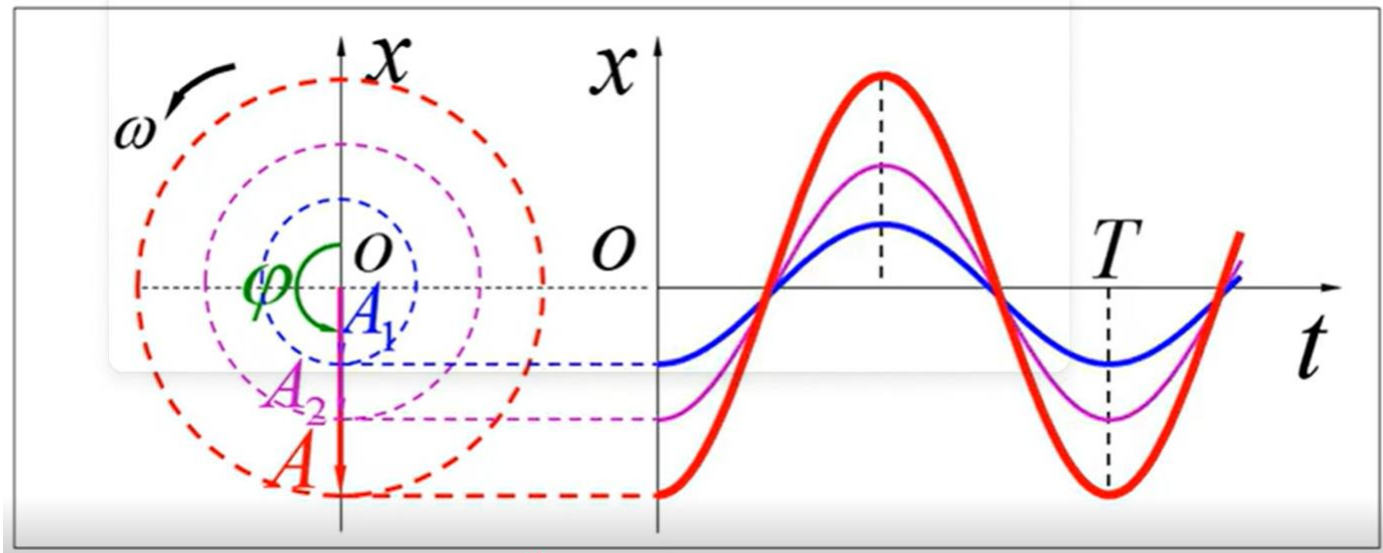
A 与 $\Delta\varphi$ 的关系

(1) 相位差 $\Delta\varphi = \varphi_2 - \varphi_1 = 2k\pi (k = 0, \pm 1, \pm 2 \dots)$

$$\varphi = \varphi_2 = \varphi_1 + 2k\pi$$

$$A = A_1 + A_2$$

$$x = (A_1 + A_2) \cos(\omega t + \varphi)$$



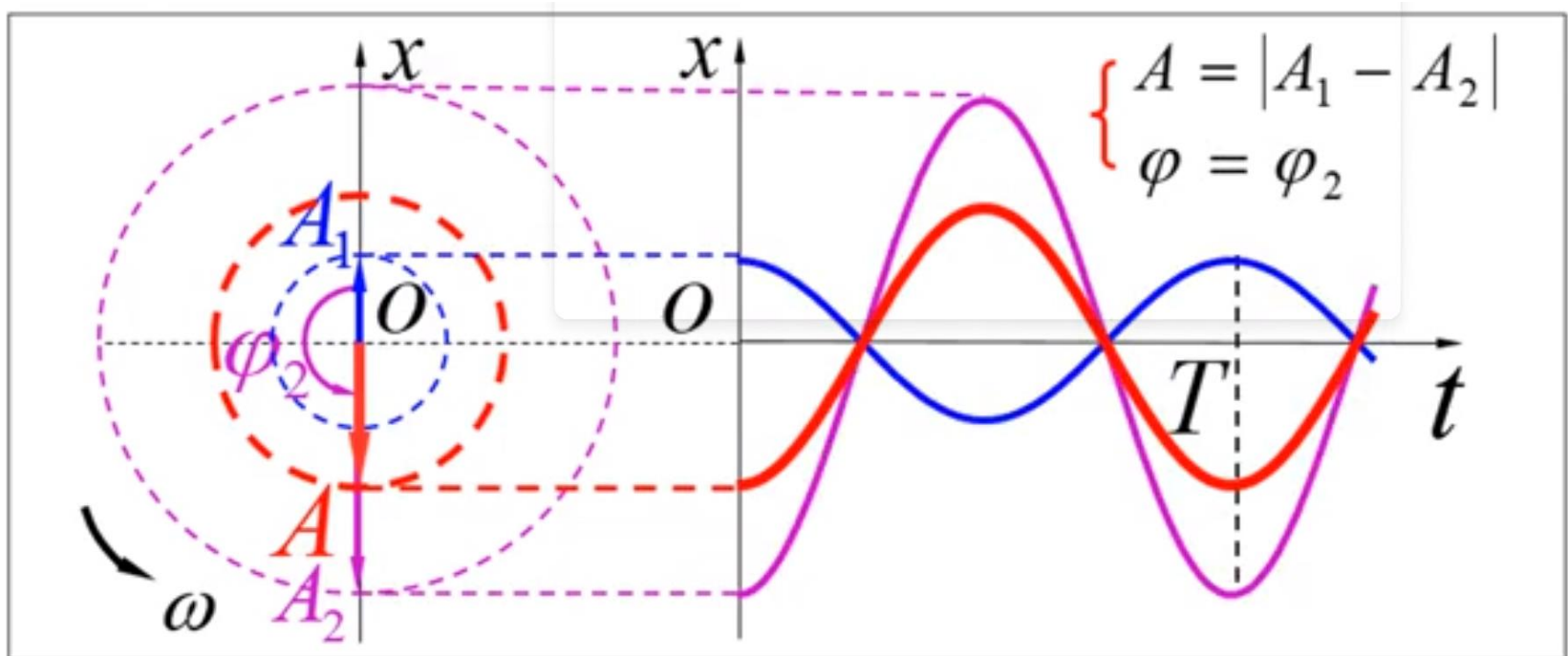
$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\varphi_2 - \varphi_1)}$$

(2) 相位差 $\Delta\varphi = \varphi_2 - \varphi_1 = (2k + 1)\pi (k = 0, \pm 1, \pm 2 \dots)$

$$x_1 = A_1 \cos \omega t$$

$$x_2 = A_2 \cos(\omega t + \pi)$$

$$x = |(A_1 - A_2)| \cos(\omega t + \varphi)$$



(1)相位差 $\Delta\varphi = \varphi_2 - \varphi_1 = 2k\pi (k = 0, \pm 1, \pm 2 \cdots)$

$$x = (A_1 + A_2) \cos(\omega t + \varphi)$$

$$A = A_1 + A_2$$

相互加强

(2)相位差 $\Delta\varphi = \varphi_2 - \varphi_1 = (2k + 1)\pi (k = 0, \pm 1, \pm 2 \cdots)$

$$x = |(A_1 - A_2)| \cos(\omega t + \varphi)$$

$$A = |A_1 - A_2|$$

相互削弱

(3)相位差 $\Delta\varphi$ 为其他值（一般情况）

$$A_1 + A_2 > A > |A_1 - A_2|$$

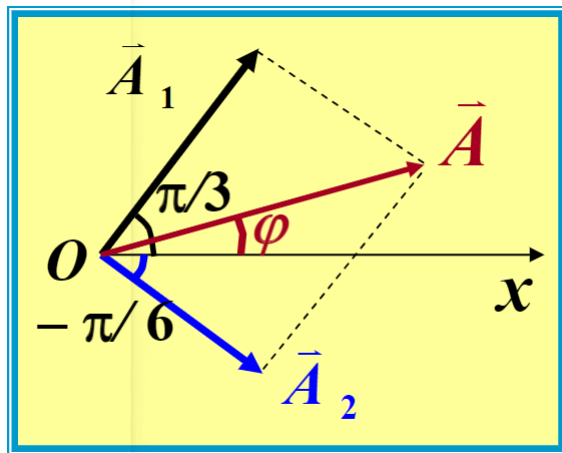
例题：某质点同时参与两个同方向、同频率的简谐振动 $x_1 = 0.4 \cos(3t + \frac{\pi}{3})$ $x_2 = 0.3 \cos(3t + \frac{\pi}{6})$

求：(1) 合振动的表达式

(2) 若 $x_3 = 0.5 \cos(3t + \varphi_3)$

φ_3 为多少时，三者合振幅最大

φ_3 为多少时，三者合振幅最小！



解：(1) $\varphi_1 - \varphi_2 = \pi / 2$ $A = \sqrt{A_1^2 + A_2^2} = 0.5(\text{m})$

$$\angle A_1 O A = \alpha \quad \tan \alpha = \frac{0.3}{0.4} \quad \alpha = 0.21\pi$$

$$\varphi = \pi / 3 - \alpha = 0.12\pi$$

$$x = 0.5 \cos(3t + 0.12\pi)$$

(2) $\varphi_3 = \varphi = 0.12\pi$ $A_{\max} = A_1 + A_2 = 0.5 + 0.5 = 1\text{m}$

(3) $\varphi_3 = \varphi + \pi = 1.12\pi$ $A_{\min} = |A_1 - A_2| = 0.5 - 0.5 = 0$

2、同方向、不同频率简谐振动的合成

$$x_1 = A_1 \cos(\omega t + \varphi_1)$$

$$x_2 = A_2 \cos(\omega t + \varphi_2)$$

$$\Delta\varphi = (\omega_2 t + \varphi_2) - (\omega_1 t + \varphi_1) = (\omega_2 - \omega_1)t + (\varphi_2 - \varphi_1)$$

合振动 $x = x_1 + x_2 = A \cos(\omega_1 t + \varphi_1) + A \cos(\omega_2 t + \varphi_2)$

若 $\varphi_1 = \varphi_2 = 0$ $x = 2A \cos\left(\frac{\omega_2 - \omega_1}{2}t\right) \cdot \cos\left(\frac{\omega_2 + \omega_1}{2}t\right)$

合振动不是简谐振动

$$x = 2A \cos\left(\frac{\omega_2 - \omega_1}{2}t\right) \cdot \cos\left(\frac{\omega_2 + \omega_1}{2}t\right)$$

特例：若 ω_2 与 ω_1 都比较大，但是数值相近

设 $\omega_2 > \omega_1$ ($v_2 > v_1$), $\omega_2 - \omega_1 \ll \omega_2 + \omega_1$

$$A(t) = 2A \cos\left(\frac{\omega_2 - \omega_1}{2}t\right) \quad \text{随}t\text{缓变}$$

$$x = A(t) \cos \varpi t$$

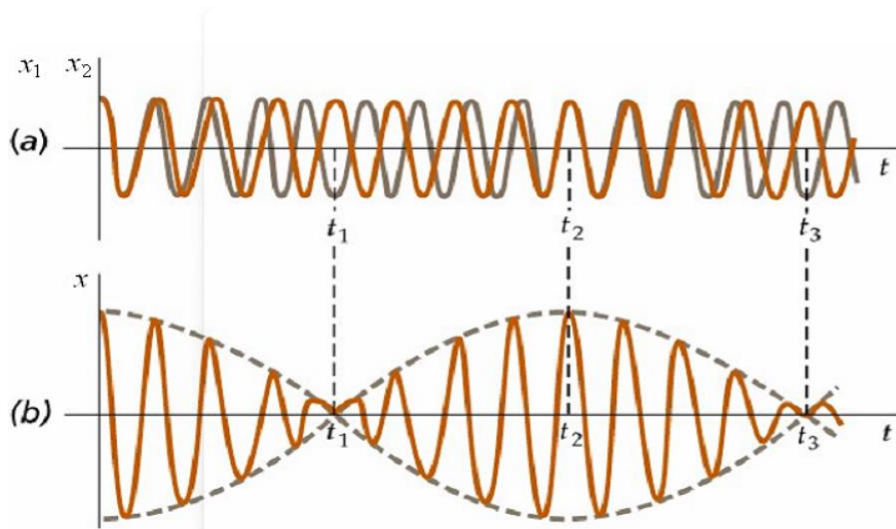
$$\cos \varpi t = \cos\left(\frac{\omega_2 + \omega_1}{2}t\right) \quad \text{随}t\text{快变}$$

合振动可看作振幅
缓变的简谐振动

$$2A \cos\left(\frac{\omega_2 - \omega_1}{2}t\right)$$

平均圆频率： $\omega_{\text{平均}} = \frac{\omega_2 + \omega_1}{2}$

调制圆频率： $\omega_{\text{调制}} = \frac{\omega_2 - \omega_1}{2}$



$$x = A(t) \cos \bar{\omega} t$$

$$A(t) = 2A \cos\left(\frac{\omega_2 - \omega_1}{2} t\right)$$

$$\cos \bar{\omega} t = \cos\left(\frac{\omega_2 + \omega_1}{2} t\right)$$

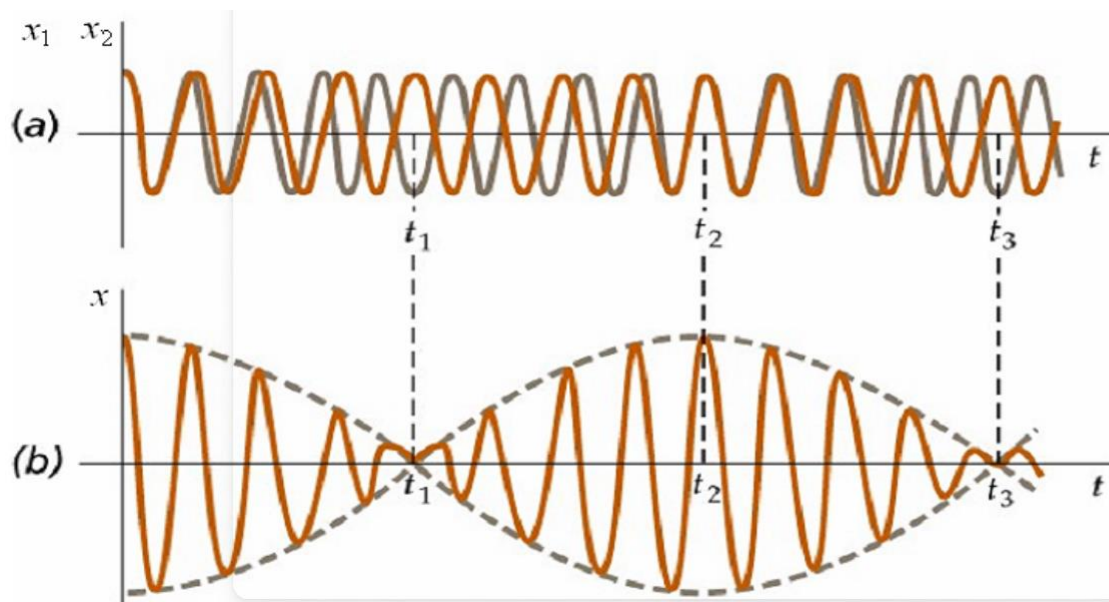
振动方向相同、频率之和远远大于频率之差的两个简谐振动合成时，合振动振幅呈周期性变化的现象称为拍。

合振幅变化一个周期为一拍。

单位时间内拍出现的次数为拍频。

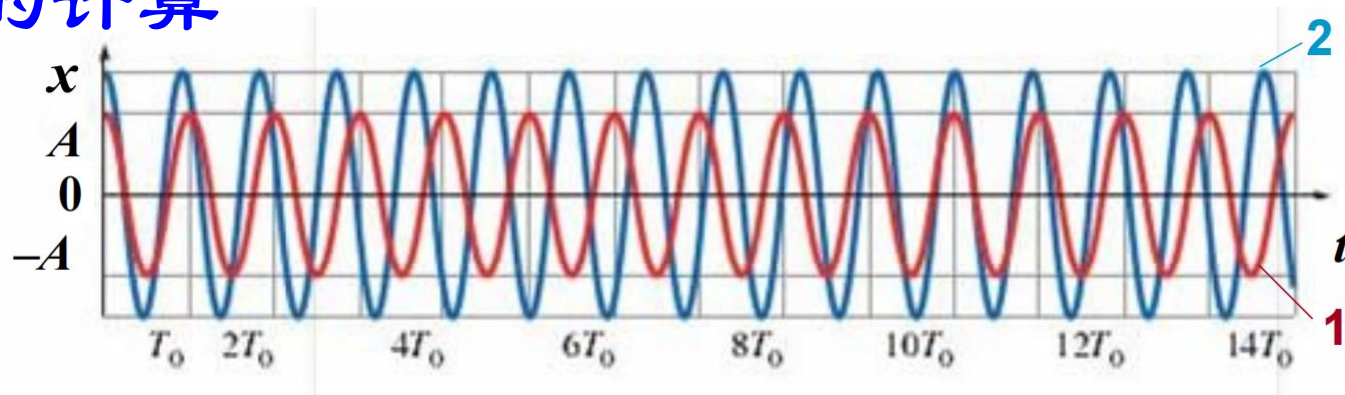
结论：同一直线上不同频率的简谐振动的合成

在 ω_2 与 ω_1 都比较大，且数值相近的条件下，合振动可视为振幅缓慢变化的简谐振动，合振动的振幅周期性变化，出现拍的现象。



拍频: $\nu = |\nu_2 - \nu_1|$

拍的计算



$$\nu_1 = 1/T_0 \quad \nu_2 = 1.1\nu_1 \quad \nu_2 = 1.1/T_0$$

$$\text{拍频: } \nu = 0.1/T_0$$

例如：钢琴调音师敲击音叉（频率为523.3Hz）

同时敲击一个钢琴琴键。两者频率近似相等，他听到了每秒有3个拍。逐渐张紧琴弦，他听到拍频慢慢降为2个拍。已知使琴弦张紧会提高弦的频率，求琴弦张紧前的频率为多少？

解： $\Delta\nu = 3\text{Hz}$ $\nu_0 = 523.3 - 3 = 520.3\text{Hz}$

相互垂直的简谐振动的合成

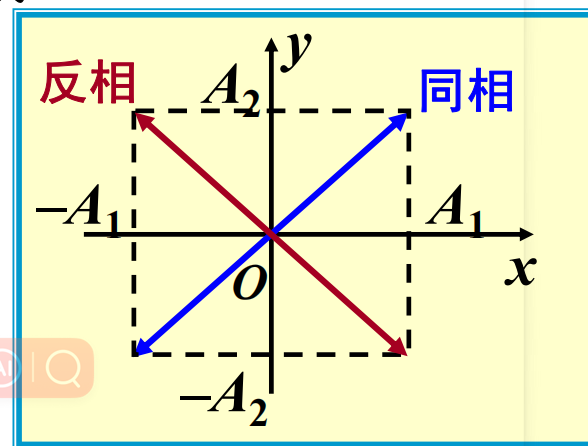
1、同周期

$$x = A_1 \cos(\omega t + \varphi_1)$$

$$y = A_2 \cos(\omega t + \varphi_2)$$

(1) $\varphi_2 - \varphi_1 = 0$ 同相 $\varphi_2 = \varphi_1 = \varphi$

$$\frac{x}{A_1} = \frac{y}{A_2} \quad y = \frac{A_2}{A_1} x$$



质点离开原点的位移 s 为: $s = \sqrt{A_1^2 + A_2^2} \cos(\omega t + \varphi)$

合振动为简谐振动

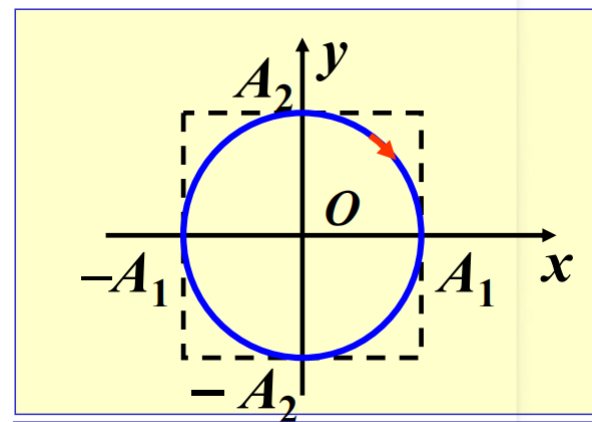
(2) $\varphi_2 - \varphi_1 = \pi$ 反相 $y = -\frac{A_2}{A_1} x$

$$x = A_1 \cos(\omega t + \varphi_1) \quad y = A_2 \cos(\omega t + \varphi_2)$$

(3) $\varphi_2 - \varphi_1 = \pi / 2$ y 超前 x

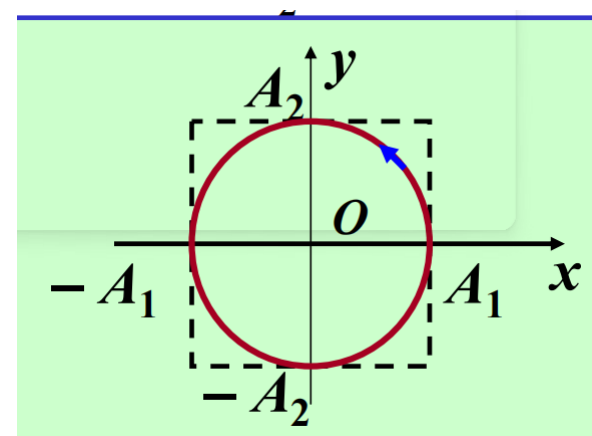
$$\frac{x_1^2}{A_1^2} + \frac{y_1^2}{A_2^2} = 1$$

轨迹：以坐标轴为轴的正椭圆
顺时针



(4) $\varphi_2 - \varphi_1 = -\pi / 2$ x 超前 y

$$\frac{x_1^2}{A_1^2} + \frac{y_1^2}{A_2^2} = 1 \quad \text{逆时针}$$



$$x = A_1 \cos(\omega t + \varphi_1)$$

$$y = A_2 \cos(\omega t + \varphi_2)$$

(5) 一般 $\Delta\varphi = \varphi_1 - \varphi_2$

$$\frac{x_1^2}{A_1^2} + \frac{y_1^2}{A_2^2} - \frac{2xy}{A_1 A_2} \cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1) \quad \text{轨迹为椭圆}$$

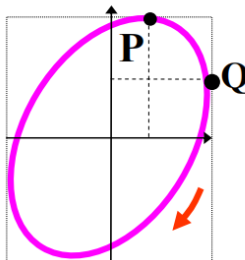
$$\Delta\varphi = \varphi_2 - \varphi_1$$

y 超前 x

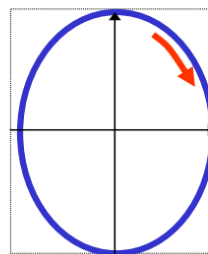
$$\Delta\varphi = 0$$



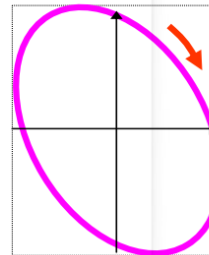
$$\Delta\varphi = \pi/4$$



$$\Delta\varphi = \pi/2$$

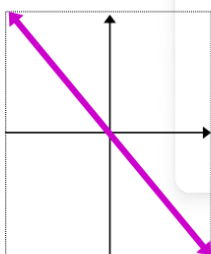


$$\Delta\varphi = 3\pi/4$$

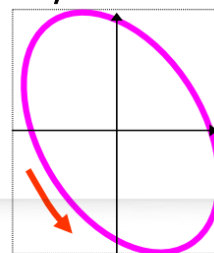


x 超前 y

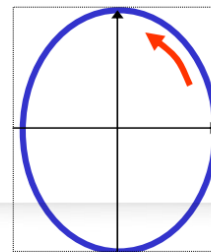
$$\Delta\varphi = \pi$$



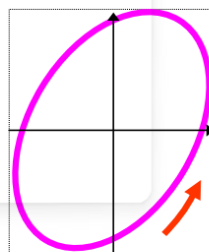
$$\Delta\varphi = 5\pi/4$$



$$\Delta\varphi = 3\pi/2$$

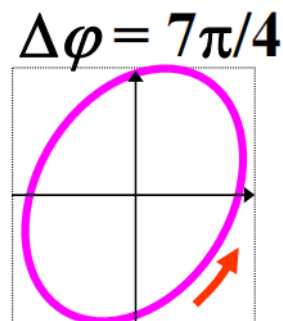
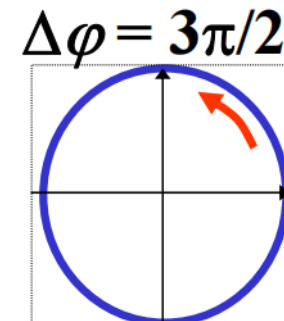
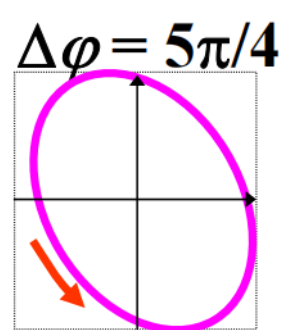
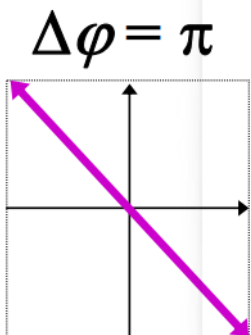
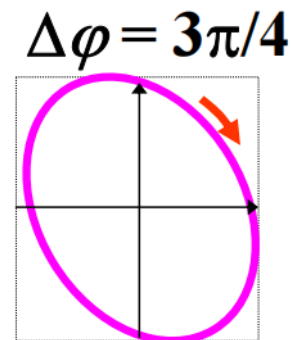
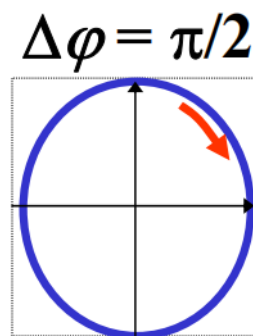
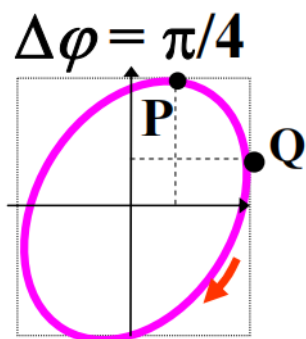
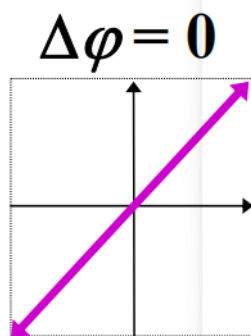


$$\Delta\varphi = 7\pi/4$$



2、相互垂直、不同周期简谐振动的合成

(1) 两振动频率相差很小



2、相互垂直、不同周期简谐振动的合成

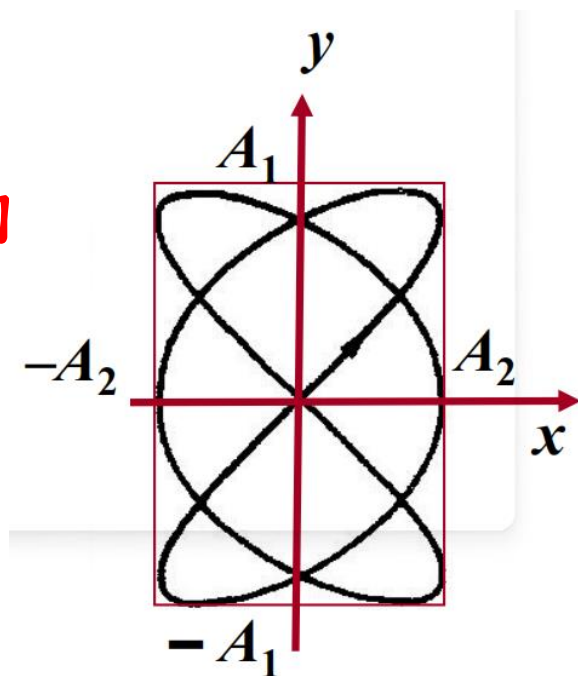
(2) 两振动频率相差很大，但有简单的整数比

$$\frac{\omega_x}{\omega_y} = \frac{m}{n}, m, n \text{ 正整数}$$

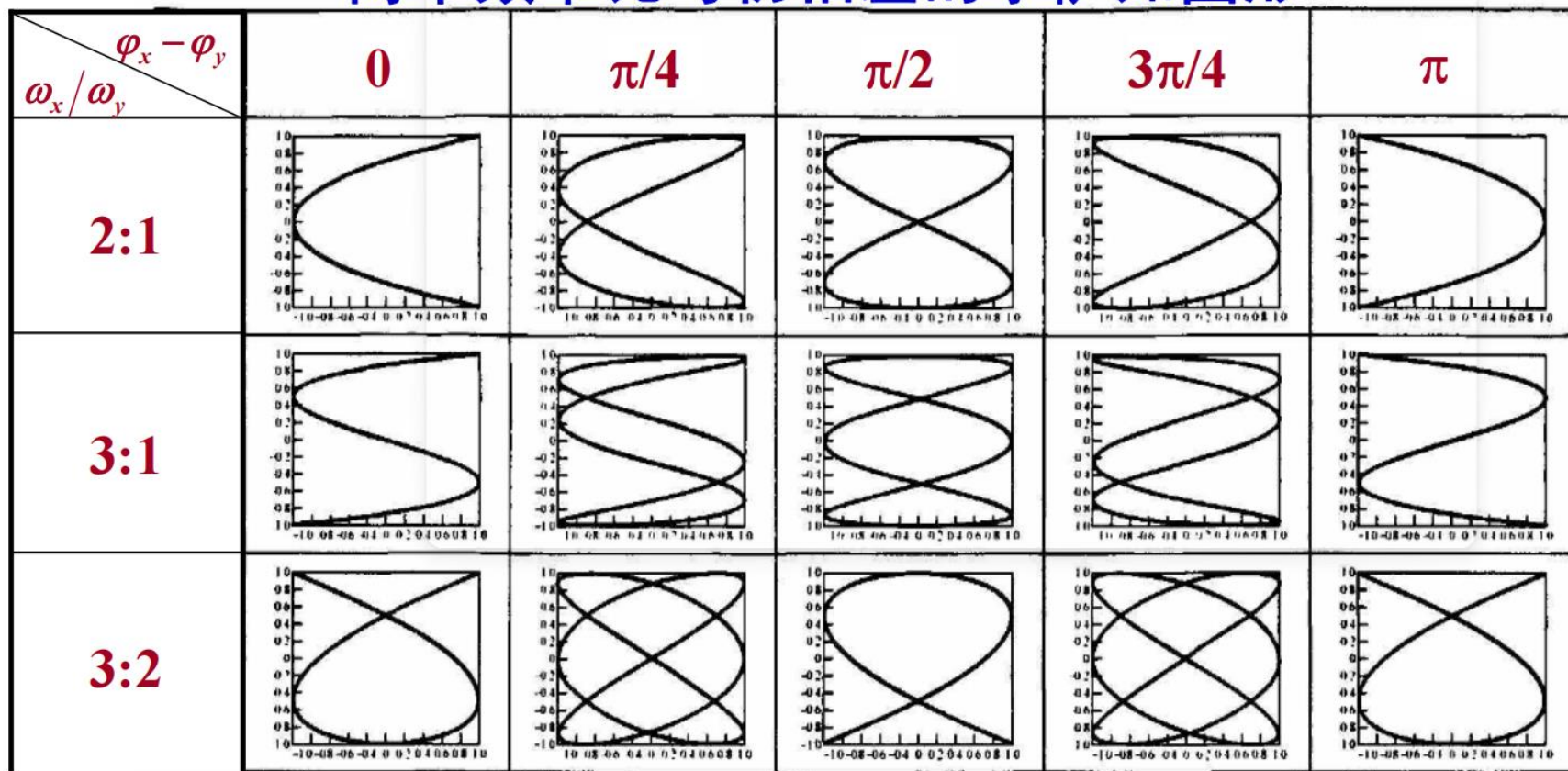
合成轨迹为稳定的曲线-李萨如图

如图：
$$\frac{\omega_x}{\omega_y} = \frac{3}{2}$$

应用：测定未知频率



简单频率比与初相差的李萨如图形



阻尼振动、受迫振动和共振

